Akshay Patil

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EDUCATION

Georgia Institute of Technology, Atlanta, GA **May 2029**
Bachelor of Science in Aerospace Engineering | Sophomore | GPA: 4.00
Current coursework: AE 1601 Intro to Aerospace Engineering, COE 2001 Statics, MSE 2001 Materials, MATH 2551 Multivariable Calculus, ME 1670 SolidWorks, CS 1371 MATLAB

School of Science and Engineering at Townview, Dallas, TX **Aug 2022 – May 2026**
3.97 Unweighted GPA | Rank 2 | 1600 SAT | National Merit Scholar | FIRST Alum

EXPERIENCE

YC-Backed Stealth Startup | Hardware Intern **Aug 2026 – Present**
- Building the world's fastest RC interceptor aircraft
- Designed the nose cone and avionics bay, replacing adhesive mounting so the airframe can be built at scale
- Designed 3D printed rib alignment jigs and built two RC test aircraft on them, drilling, finishing, and epoxy bonding carbon fiber structure
- Responsible engineer for the next two programs: a subscale supersonic demonstrator and a fully 3D printed jet interceptor

Hardpoint Robotics | Co-Founder **Jun 2026 – Present**
- Break-even startup building NDAA-compliant development UAVs for customers barred from foreign hardware
- Modeled a 500 mm, 4 kg max takeoff weight quadcopter in CAD with 3D printed CF-Nylon parts, sized for 12.7 min endurance with a 410 g Jetson and stereo camera payload
- Working with North American vendors to manufacture carbon fiber structures and source compliant components, negotiating pricing and delivery timelines

SWYFT Robotics | Engineering Design Intern **May 2025 – May 2026**
- SWYFT sells competition drivetrain and structural parts to FIRST Tech Challenge and FIRST Robotics Competition teams
- Designed sprockets, gears, and pulleys for competition loads, generating $10,000 in revenue
- Created a modular robot cart for FRC teams with a detachable programming station, designed for extrusion and fiber laser cutting

RESEARCH AND CLUBS

Hypersonics VIP, Georgia Tech | Data and Communications Team **Aug 2026 – Present**
- Building a sonic fatigue test facility, where acoustic loading is used to find how structures fail under sustained high-intensity sound
- Designing the sensor package that measures pressure, temperature, and airflow inside the facility
- Developing parabolic mirrors to generate a uniform sound field across the test section

Yellow Jacket Space Program | Turbomachinery, Engine Development **Sep 2026 – Present**
- Pursuing the collegiate liquid propellant altitude record and the Karman line at 100 km
- Designing electric pumps and turbopumps so the vehicle can drop the heavy pressurant tanks; drew the GD&T part drawings for the Elytra electric pump
- Simulated the Vespula engine's LOx injector manifold in STAR-CCM+, showing all 17 injectors run within 1.1% of mean flow and tracing the remaining imbalance to injector ring layout

FIRST Tech Challenge Team 7172 (Top 0.1% Worldwide) | Captain and Lead Designer **Sep 2021 – May 2026**
- Lead designer on four robots; 2026 FIRST World Championship Finalist Alliance, top 8 of roughly 7,000 teams
- Designed mechanisms using linkages, counter-springing, pulley systems, and geartrains
- Used industrial arm kinematics to design and build a 6-DoF arm robot that expands from an 18 in cube to 13 ft of reach
- 24 competition wins (2021 to 2026), plus Inspire, Think, Connect, Innovate, Design, Motivate, and Control awards

Beaver Works Summer Institute at MIT | UAS-SAR Track **Jun 2024 – Aug 2024**
- Built a UAV synthetic aperture radar by mounting a PulsON 440 ultra-wideband radar on a DJI F550
- Wrote a vectorized Python back projection algorithm that resolved targets even in noisy conditions

SKILLS

CAD and CFD: SolidWorks, Fusion 360, Onshape, Autodesk Inventor, GD&T drawings, STAR-CCM+
Manufacturing: CNC machining, 3D printing (CF-Nylon), carbon fiber, epoxy, jigs and fixtures
Programming: Python (NumPy, Matplotlib, Pandas, TensorFlow), MATLAB, Java, Git
Hardware: NVIDIA Jetson, Raspberry Pi, Arduino, REV Control Hub
Certifications: FAA Part 107 Remote Pilot, Autodesk Certified User (Inventor)